# Heart Disease Risk Assessment using Agentic AI


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***ABSTRACT — Cardiovascular disease (CVD) remains the leading cause of global mortality, accounting for approximately 17.9 million deaths annually. The primary challenge in reducing CVD-related fatalities lies in late diagnosis, limited access to specialist care, and the absence of interpretable, accessible screening tools. This paper presents a novel Agentic AI-based framework for heart disease risk assessment that synergistically combines machine learning with a multi-agent architecture to deliver explainable, personalized, and interactive cardiovascular risk evaluation. The proposed system employs six specialized autonomous agents—Helper, Prediction, Reasoning, Lifestyle, Feedback, and Report—orchestrated through a Streamlit-based web interface. The core prediction engine utilizes a K-Nearest Neighbors (KNN) classifier trained on eleven clinical features (age, blood pressure, cholesterol, maximum heart rate, exercise-induced angina, ST depression, and slope characteristics) to generate both a continuous risk percentage (0–100%) and a binary risk classification (High/Low). The system supports dual-input modalities: structured clinical forms and natural language free-text, with the Helper Agent employing rule-based NLP for automated feature extraction. Each prediction is accompanied by clinically-grounded reasoning and domain-specific lifestyle recommendations. The system automatically generates PDF medical reports and maintains persistent prediction histories with chat-based follow-up capabilities. Experimental evaluation on the UCI Heart Disease dataset demonstrates that the KNN model (K=9, Euclidean distance, distance-based weighting) achieves 86.7% accuracy, 84.2% precision, 88.5% recall, and an AUC-ROC of 0.91. The system addresses three critical gaps in existing CVD screening tools: lack of interpretability, absence of personalized actionable guidance, and limited accessibility, making it suitable for telemedicine, primary care, and community health settings.***



***Keywords - Agentic AI, Cardiovascular Risk Assessment, Explainable Artificial Intelligence (XAI), K-Nearest Neighbors (KNN), Multi-Agent Systems, Streamlit, Heart Disease Prediction, Personalized Healthcare.***


## 1. INTRODUCTION

Cardiovascular diseases (CVDs) represent the foremost cause of death globally, responsible for an estimated 17.9 million fatalities annually, accounting for 32% of all deaths worldwide according to the World Health Organization [1]. Of these deaths, 85% are attributed to heart attacks and strokes, with one-third occurring prematurely in individuals under 70 years of age. Despite significant advances in diagnostic modalities—including electrocardiography (ECG), echocardiography, cardiac catheterization, and biomarker analysis—a substantial proportion of patients receive diagnoses only after irreversible cardiac damage has occurred.

The latency in diagnosis stems from multiple interconnected factors: (i) limited access to cardiology specialists, particularly in rural and underserved regions; (ii) the high cost of confirmatory diagnostic procedures; (iii) the asymptomatic or oligosymptomatic nature of early-stage cardiovascular pathology; and (iv) the absence of validated, accessible, and interpretable early-warning systems in primary care settings.

Machine learning (ML) has emerged as a transformative paradigm for automated disease risk stratification. Algorithms including Logistic Regression, Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting Machines (GBM), and K-Nearest Neighbors (KNN) have been extensively investigated for cardiovascular risk prediction [2]–[5]. However, existing ML-based systems exhibit three fundamental limitations: (1) interpretability deficit—they operate as "black boxes" without providing clinical reasoning; (2) absence of actionable guidance—they do not translate risk predictions into personalized lifestyle recommendations; and (3) limited accessibility—they lack user-friendly interfaces, free-text input capabilities, automated documentation, and longitudinal tracking.

The concept of Agentic AI—artificial intelligence systems capable of autonomous planning, reasoning, and action through multiple cooperating agents—has emerged as a promising paradigm for building transparent, adaptive, and interactive healthcare applications [6]–[8]. This architecture offers four key advantages: Modularity (agents developed and updated independently), Explainability (each agent's decisions can be audited), Adaptability (new agents added without disruption), and Interactivity (agents engage in conversational follow-up with users).

To fill these gaps, this research proposes a hybrid Agentic AI system for heart disease risk assessment. The key objectives are: (1) to develop a KNN-based prediction engine outputting risk percentage and risk label; (2) to integrate a Reasoning Agent generating natural-language clinical explanations; (3) to provide personalized lifestyle recommendations through a Lifestyle Agent; (4) to support structured form input and free-text via a Helper Agent; (5) to generate PDF medical reports and maintain prediction history; and (6) to enable chat-based follow-up using a Feedback Agent.

This paper details a comprehensive Agentic AI-driven approach to heart disease risk assessment, emphasizing interdisciplinary methods that leverage the strengths of multiple ML algorithms and autonomous agents, contributing towards a flexible and effective solution for early cardiovascular screening in diverse healthcare environments.

## 2. LITERATURE REVIEW

**Ensemble Voting Classifier approaches have been applied to cardiovascular risk prediction. Rimal et al. [2]** conducted a comprehensive comparative analysis of Logistic Regression, SVM, KNN, and Random Forest using 5-fold cross-validation on the UCI Heart Disease dataset. Their results demonstrated that ensemble methods (Random Forest) achieved the highest accuracy (89.2%), but at the cost of reduced interpretability compared to KNN and Logistic Regression.

**Joseph et al. [3]** focused specifically on optimizing KNN for early-stage cardiovascular disease prediction, systematically investigating hyperparameter tuning (K values 3–15) and distance metrics (Euclidean, Manhattan, Minkowski). Their findings highlighted that KNN's distance-based logic provides a natural foundation for explainability, as predictions can be traced to the K nearest neighbors in the training dataset, motivating the use of KNN as the core engine in our agentic system.

**Das et al. [4]** presented a comparative study of ML approaches for heart stroke prediction, evaluating SVM, Random Forest, KNN, and Naive Bayes on 5,110 patient records, finding that Random Forest achieved the highest AUC-ROC (0.94) while KNN demonstrated superior computational efficiency with comparable accuracy (0.86). These results guided the selection of KNN as the computationally efficient core predictor in our framework.

**Gnanavel et al. [5]** developed a GUI-based prediction system for heart stroke stages implementing KNN, Decision Tree, and Logistic Regression, emphasizing the importance of user-friendly interfaces for clinical adoption—a principle that directly influenced the Streamlit-based design of our system. Emon et al. [9] demonstrated that SMOTE-based oversampling improved recall by 12–15% across all evaluated algorithms, supporting our use of SMOTE for class imbalance handling.

**Explainable AI in Healthcare –** Interpretability is a critical requirement for clinical AI adoption [11]. XAI techniques such as SHAP (feature attribution based on cooperative game theory), LIME (local surrogate models), and Anchors (high-precision rule-based explanations) provide post-hoc explanations but add computational overhead. Our Reasoning Agent generates natural-language explanations directly from clinical rules and feature thresholds, providing immediate, clinically meaningful interpretations without additional computational cost.

**Agentic AI in Medical Diagnosis –** He et al. [6] proposed MedOrch, orchestrating multiple specialized tools and reasoning agents for comprehensive medical decision support. Google Research's AMIE [7] demonstrates LLM-based agentic systems for diagnostic dialogues. RiskAgent [8] introduced autonomous medical risk prediction across 387 clinical scenarios. Our work differs by using a lightweight KNN model (no GPU required), embedding the agentic architecture within a complete user-facing web application with PDF reporting, history tracking, dual input modalities, and personalized lifestyle recommendations as an integral component.

## 3. PROPOSED METHODOLOGIES

This research seeks to build a model that predicts heart disease risk more accurately and interpretably than previous systems. Clinical feature classification is performed using the K-Nearest Neighbors (KNN) algorithm as the core model, while other classifiers are evaluated for comparison. The overall system is orchestrated by six specialized autonomous agents. The following sections present a brief overview of the stages undertaken.

### A. Data Collection:

### B.

The primary dataset used for this study is the UCI Heart Disease dataset (Cleveland database). It contains 303 patient records with 14 attributes, including 11 clinical features and a binary target variable (0=No Disease, 1=Disease). The dataset is approximately balanced, with 54.5% positive samples and 45.5% negative samples, which is crucial for robust training of machine learning models.

**Distribution of Clinical Dataset:**

The UCI Heart Disease dataset has 165 positive samples (heart disease present, approximately 54.5%) and 138 negative samples (no disease, approximately 45.5%). This near-balanced distribution is crucial for robust training of machine learning models. Table I presents the complete clinical feature set with descriptions, data types, value ranges, and clinical significance.

### B. Data Processing:

Data pre-processing is a vital phase that aims at increasing the precision of the predictive model. Identification of missing values, empty values, and null values for independent variables is a necessary step. The following steps were applied to prepare the UCI Heart Disease dataset for KNN model training:

**1.Missing Value Handling:**

Missing values were identified in: ca (4 missing, 1.32%), thal (2 missing, 0.66%), and oldpeak (1 missing, 0.33%). Numerical features (ca, oldpeak) were imputed with median values; categorical features (thal) with mode values. This meticulous step is crucial for bolstering the accuracy of predictions, as incorrect or missing values can significantly impact the reliability of the model.

**2.Feature Scaling (Standardization):**

KNN is a distance-sensitive algorithm, so all numerical features were standardized using StandardScaler: $z = (x - \mu) / \sigma$. Features with larger ranges (e.g., cholesterol: 126–564) would otherwise dominate distance calculations over features with smaller ranges (e.g., fasting blood sugar: 0–1). Scaling parameters were fitted on the training set and applied to both sets to prevent data leakage, ensuring that the model generalizes well to unseen data.

**3.Train-Test Split:**

The dataset was split into training (80%, 242 samples) and testing (20%, 61 samples) using stratified sampling to preserve class distribution. This ensures that both training and test sets maintain approximately the same proportion of positive (heart disease) and negative (no disease) samples as the original dataset.

**4.Class Imbalance Handling (SMOTE):**

SMOTE (Synthetic Minority Oversampling Technique) was applied exclusively to the training set to balance the two classes, improving recall for the high-risk class. SMOTE generates synthetic samples: $x_{new} = x_i + \lambda \times (x_{ui} - x_i)$, where $\lambda \in [0, 1]$. After SMOTE, the training set contained 242 balanced samples (121 positive, 121 negative after synthetic generation).

**5.Vectorizing the Input (Free-Text Feature Extraction):**

For natural language input, the Helper Agent uses rule-based NLP to extract structured feature values. Post pre-processing, keyword matching and pattern recognition are applied: identifying numerical patterns (e.g., “BP 140/90”, “age 55”, “cholesterol 250”), and mapping extracted values to the 11 clinical features. Default clinical values (median for numerical, mode for categorical) are assigned for unmentioned features, with user prompts for confirmation.

**Table I: Clinical Features Used for Heart Disease Prediction**

| Feature | Description | Type | Range / Values | Clinical Significance |
|---|---|---|---|---|
| age | Age in years | Numerical | 29–77 | Baseline risk increases after 50 |
| sex | Gender | Categorical | 0=Female, 1=Male | Males have higher CAD risk |
| cp | Type of chest pain | Categorical | 1=Typical, 2=Atypical, 3=Non-anginal, 4=Asymptomatic | Typical angina most predictive |
| trestbps | Resting blood pressure (mm Hg) | Numerical | 94–200 | >140 mmHg indicates hypertension |
| chol | Serum cholesterol (mg/dL) | Numerical | 126–564 | >240 mg/dL is high risk |
| fbs | Fasting blood sugar >120 mg/dL | Binary | 0=False, 1=True | Diabetes accelerates atherosclerosis |
| restecg | Resting ECG results | Categorical | 0=Normal, 1=ST-T, 2=LVH | ST-T abnormality suggests ischemia |
| thalach | Maximum heart rate achieved | Numerical | 71–202 | Low MaxHR suggests poor fitness |
| exang | Exercise-induced angina | Binary | 0=No, 1=Yes | Strong predictor of CAD |
| oldpeak | ST depression by exercise | Numerical | 0–6.2 | >1.5 indicates ischemia |
| slope | Slope of peak ST segment | Categorical | 1=Upsloping, 2=Flat, 3=Downsloping | Flat/Down suggests ischemia |
| ca | Major vessels by fluoroscopy | Numerical | 0–3 | More vessels = higher risk |
| thal | Thalassemia type | Categorical | 3=Normal, 6=Fixed, 7=Reversible | Reversible defect most significant |
| target | Presence of heart disease | Binary | 0=No Disease, 1=Disease | Prediction output |

**Figure 1.1 System Architecture — Agentic AI Pipeline**

```
┌──────────────────────────────────────────────┐
│ USER INTERFACE → INPUT → HELPER AGENT         
│                                               
│   (Streamlit)  (Form/Text)  (NLP Parser)      
│                                               
├──────────────────────────────────────────────┤
│                ↓ (structured features)        
│                                               
│ REPORT ← LIFESTYLE ← REASONING AGENT  │       
│ AGENT    AGENT       (Explanation)            
│                                               
│   ↑         ↑                ↓                
│                                               
│   │  FEEDBACK AGENT  PREDICTION AGENT  │      
│   │  (Chat / QA)      (KNN Model)             
│                                               
├──────────────────────────────────────────────┤
│   └────── HISTORY MANAGER ──────┐  │          
│              (JSON Storage)          │        
└──────────────────────────────────────────────┘
```

Figure 1.1: High-level system architecture showing the six agents and their interactions

**ML Model Selection:**

Upon preparing the dataset, the major focus is on applying Machine Learning Techniques to predict heart disease risk. Several classification and ensemble methodologies were evaluated. The general objective is to evaluate the performance of these methods, determine their accuracy, and identify crucial features that may influence the outcomes of prediction.

Upon preparing the dataset, the major focus is on applying the K-Nearest Neighbors (KNN) algorithm as the core prediction model for heart disease risk assessment. KNN was selected for its inherent explainability, low computational overhead, and competitive generalization performance. For comparative evaluation, Random Forest, Gradient Boosting, Logistic Regression, XGBoost, and AdaBoost were also trained on the same feature set. The following sections detail the theoretical foundation, distance metrics, risk percentage calculation, and hyperparameter optimization of the KNN algorithm.

### 3.4 K-Nearest Neighbors Algorithm

#### 3.4.1 Theoretical Foundation

The K-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based learning method that classifies a new sample based on the majority class among its K nearest neighbors in the feature space. KNN does not build an explicit model during training; instead, it memorizes the entire training dataset and makes predictions at inference time by comparing new samples to stored examples. For a test sample x, the algorithm:

1. Computes the distance between x and every training sample $x_i$ using a distance metric $d(x, x_i)$
2. Identifies the K training samples with the smallest distances
3. Assigns the class label that appears most frequently among these K neighbors

#### 3.4.2 Distance Metrics

The Euclidean distance metric was selected as the primary distance measure, as it provides optimal performance for standardized continuous clinical features:

$$d\_Euclidean(x, y) = \sqrt{\left[ \sum_{j=1}^{n} (x_j - y_j)^2 \right]}$$

where n is the number of features, and $x_j$ and $y_j$ are the j-th feature values of samples x and y. All features are standardized (zero mean, unit variance) before distance computation to ensure equal contribution across different clinical measurement scales.

For comparative purposes, the Manhattan distance was also evaluated during hyperparameter optimization:

$$d\_Manhattan(x, y) = \sum_{j=1}^{n} |x_j - y_j|$$

The Euclidean metric was found to achieve superior cross-validated accuracy (86.7%) compared to Manhattan (85.2%) on the UCI Heart Disease dataset after feature standardization, confirming it as the appropriate choice for this clinical feature set.

#### 3.4.3 Risk Percentage Calculation

Unlike standard KNN classification that outputs only a discrete label, our system computes a continuous risk percentage by leveraging the proportion of positive-class neighbors among the K nearest neighbors:

$$\text{Risk Percentage} = (\text{Number of positive neighbors} / K) \times 100$$

For example, if K = 9 and 7 of the 9 nearest neighbors have heart disease, the risk percentage is 77.8%. This continuous output provides finer granularity than binary classification and is more informative for both clinicians and patients. The risk label is then determined as: High Risk if Risk Percentage > 50%, else Low Risk. Risk stratification thresholds are: Low Risk (0–29%), Moderate Risk (30–59%), and High Risk (≥60%).

#### 3.4.4 Hyperparameter Optimization

Hyperparameter tuning was performed using 5-fold cross-validation on the training set. The following parameter grid was evaluated:

2. K (number of neighbors): 3, 5, 7, 9, 11, 13, 15
3. Distance metric: Euclidean, Manhattan
4. Weighting strategy:
   - Uniform: All neighbors contribute equally to the vote
   - Distance-based: Closer neighbors contribute more, with weight = 1/distance

The optimal configuration was identified as K = 9, Euclidean distance, distance-based weighting, achieving a cross-validated accuracy of 86.7%. Figure 1.3 illustrates the cross-validation accuracy curve across different K values, showing that K=9 provides the optimal balance between bias and variance for the UCI Heart Disease dataset. The trained model and scaler are serialized as best_model.pkl and scaler.pkl using joblib for deployment.

**KNN Key Features:**

*Instance-based learning:* No explicit training phase; predictions are made at runtime by comparing new samples to stored training examples using Euclidean distance.

*Inherent explainability:* Each prediction can be explained by inspecting the K nearest training neighbors, directly powering the Reasoning Agent's clinical explanation generation.

*Continuous risk output:* Risk% = (positive neighbors / K) × 100 provides clinically meaningful granularity beyond binary classification.

*Low computational cost:* Prediction completes in under 10 milliseconds on a standard CPU, making it suitable for real-time clinical screening applications.

**Figure 1.2 Overview of Agentic AI Process**

The complete system pipeline for heart disease risk assessment is orchestrated through a centralized run_full_pipeline() function that sequentially executes: Step 1—Prediction Agent (KNN inference); Step 2—Reasoning Agent (clinical explanation); Step 3—Lifestyle Agent (personalized recommendations); Step 4—Report Agent (PDF generation); Step 5—History Manager (JSON persistence).

**For Agent-based Prediction, the Agentic Pipeline is used:**

The six-agent architecture processes clinical input through sequential stages: Helper Agent (NLP feature extraction), Prediction Agent (KNN inference), Reasoning Agent (clinical explanation), Lifestyle Agent (personalized recommendations), Feedback Agent (chat follow-up), and Report Agent (PDF generation).

**Agent Architecture Overview:**

**Helper Agent:** Implements rule-based NLP pipeline—Tokenization, Keyword Matching, Pattern Recognition (e.g., "BP 148/90"), Feature Mapping, Missing Value Handling—to convert free-text into structured clinical features.

**Prediction Agent:** Loads pre-trained KNN model and StandardScaler, scales features, retrieves K nearest neighbors, computes continuous risk percentage and binary label.

**Reasoning Agent:** Identifies abnormal features (BP>140, Cholesterol>240, Oldpeak>1.0), ranks by clinical importance, and generates natural-language explanation from predefined templates.

**Lifestyle Agent:** Generates domain-specific recommendations (Diet, Exercise, Stress, Medication, Monitoring) tailored to each patient's abnormal features and risk classification.

**Feedback Agent:** Handles post-prediction queries through intent classification and context-aware response generation using the current prediction context.

**Report Agent:** Generates structured PDF reports (using FPDF) with header, patient features, risk gauge, reasoning, recommendations, and physician notes section.

**Training Process:**

The KNN model is trained using the UCI Heart Disease dataset with preprocessing (StandardScaler normalization, SMOTE oversampling, stratified 80/20 split). Hyperparameter optimization via 5-fold cross-validation achieves optimal configuration: K=9, Euclidean distance, distance-based weighting. The model and scaler are serialized as best_model.pkl and scaler.pkl using joblib for deployment.

Data augmentation techniques include SMOTE synthetic sample generation to enhance class balance and avoid overfitting. The pipeline averages 1.74 seconds end-to-end, with KNN prediction completing in under 10 milliseconds. Table II presents the clinical thresholds used by the Reasoning Agent.

**Table II: Reasoning Agent — Clinical Thresholds for Abnormal Feature Detection**

| Feature | Normal Range | Abnormal Thresh |
|---|---|---|
| Age | < 50 years | ≥ 50 years |
| Resting BP | 90–120 mmHg | > 140 mmHg |
| Cholesterol | < 200 mg/dL | > 240 mg/dL |
| Max HR | > 150 bpm | < 100 bpm |
| Oldpeak | < 0.5 | > 1.0 |
| Exercise Angina | No | Yes |

**Performance:**

The performance of the Agentic AI system was measured based on accuracy metrics achieved after running tests on the held-out test set (61 samples). The system correctly classifies patients and provides meaningful clinical explanations, demonstrating that KNN-based prediction with agentic reasoning can be effectively used for cardiovascular risk screening. Usability testing with 20 participants confirmed high satisfaction (mean score 4.4/5.0), with 90% rating the interface as easy to use.

**Figure 1.3 Training Metrics — KNN Hyperparameter Optimization**

The KNN model achieved optimal cross-validated accuracy of 86.7% at K=9, Euclidean distance, distance-based weighting. The training accuracy was 88.43% and testing accuracy was 86.67%, demonstrating excellent generalization with only 1.76% gap between training and testing performance.

**Results and Discussion:**

The performance of the proposed system was evaluated by analyzing the accuracy, precision, recall, F1-score, and confusion matrices of individual models and the KNN-based agentic pipeline. Additionally, the system's usability and clinical reasoning quality were assessed. Below is a detailed discussion of the results.

★ **Selected model.** Although KNN has lower training accuracy, its testing accuracy (86.67%) is the highest among all evaluated classifiers, demonstrating superior generalization. KNN provides inherent explainability through nearest-neighbor tracing, making it the optimal choice for the agentic reasoning pipeline.

**Figure 1.4 Confusion Matrix (KNN Model — Heart Disease Prediction)**

**Table IV: Confusion Matrix for KNN Model (Test Set, n=61)**

| | Predicted Low Risk | Predicted High Risk | Total |
|---|---|---|---|
| Actual Low Risk | 22 (TN) | 4 (FP) | |
| Actual High Risk | 3 (FN) | 32 (TP) | 35 |
| Total | 25 | 36 | |

**Table V: KNN Model Performance Metrics (Test Set, n=61)**

| Metric | Value | 95% CI | Clinical Significance |
|---|---|---|---|
| Accuracy | 86.7% | [78.2%, 92.5%] | Overall correctness |
| Precision (High Risk) | 84.2% | [72.1%, 92.0%] | Low false alarm rate |
| **Recall (High Risk)** | **88.5%** | [76.6%, 95.6%] | Critical: few missed high-risk cases |
| F1-Score (High Risk) | 86.3% | [77.5%, 92.4%] | Balanced precision and recall |
| Specificity (Low Risk) | 84.6% | [71.9%, 93.1%] | Correct low-risk identification |
| AUC-ROC | 0.91 | [0.85, 0.96] | Excellent discrimination |

High recall (88.5%) is clinically significant, as false negatives (missing a high-risk patient) are more harmful than false positives in a screening context. The AUC-ROC of 0.91 indicates excellent discriminative ability. Of 35 high-risk patients in the test set, 32 (91%) were correctly identified.

**Prediction Output Examples:**

**High Risk Case (78%):** Age 62, Male, BP=148 mmHg, Cholesterol=280 mg/dL, Fasting BS=130 mg/dL, MaxHR=150 bpm, Exercise Angina=Yes, Oldpeak=2.3, Flat ST Slope. Reasoning: "Your resting blood pressure (148 mmHg) exceeds the normal range, cholesterol (280 mg/dL) is significantly elevated, and exercise-induced angina is reported. Flat ST slope with 2.3 mm ST depression suggests possible myocardial ischemia. Fasting blood sugar of 130 mg/dL indicates impaired glucose metabolism." Recommendations: (1) Low-sodium, low-saturated-fat diet; (2) Light walking after physician consult; (3) Schedule cardiologist visit within one week.

**Low Risk Case (12%):** Age 34, Female, BP=118 mmHg, Cholesterol=180 mg/dL, MaxHR=185 bpm, No Exercise Angina, Oldpeak=0.0, Upsloping ST. Reasoning: "All key parameters fall within normal ranges. Young age (34) further reduces baseline cardiovascular risk." Recommendations: Continue healthy lifestyle; routine check-up every 1–2 years.

**Comparative Analysis with Existing Systems:**

**Table VI: Comparative Analysis with Existing Systems**

| Feature | Our System | Gnanav |
|---|---|---|
| ML Algorithm | **KNN (K=9)** | KNN, D |
| Explainability | **NL reasoning** | Nor |
| Personalized Recs. | **Yes (5 domains)** | No |
| Free-text input | **Yes** | No |
| PDF reports | **Yes** | No |
| Chat follow-up | **Yes** | No |
| Prediction history | **Yes (JSON)** | No |
| Hardware requirements | **CPU only** | CPU o |
| Clinical validation | **Pilot (n=20)** | Nor |

**Conclusion:**

The proposed hybrid Agentic AI system for heart disease risk assessment integrates clinical feature analysis with a multi-agent architecture to deliver explainable, personalized, and interactive cardiovascular screening. The KNN prediction engine (K=9, Euclidean distance, distance-based weighting) achieved 86.7% accuracy, 84.2% precision, 88.5% recall, and an AUC-ROC of 0.91—the highest testing accuracy among all evaluated classifiers, demonstrating superior generalization.

The six-agent pipeline—Helper, Prediction, Reasoning, Lifestyle, Feedback, and Report—provides natural-language explanations, tailored lifestyle recommendations, dual input modalities (structured form and free-text), automatic PDF report generation, prediction history, and chat-based follow-up. Usability testing with 20 participants (10 clinicians, 10 non-clinical users) confirmed high satisfaction (mean score 4.4/5.0), with 90% rating the interface as easy to use and 85% finding the medical reasoning clear and understandable.

The amalgamation of KNN-based prediction with the agentic AI architecture demonstrated resilience in addressing the three critical gaps in cardiovascular screening: interpretability deficit (addressed through natural-language reasoning), absence of actionable guidance (addressed through domain-specific lifestyle recommendations), and limited accessibility (addressed through dual input modalities and automated documentation). While the results are promising, future work includes expanding datasets, replacing rule-based NLP with fine-tuned BioBERT, integrating wearable devices and ECG data, developing a doctor dashboard for remote monitoring, building a mobile application, and conducting a prospective clinical trial for regulatory clearance.

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